

Ex.no:7b	DATA VISUALIZATION USING TABLEAU- HOTEL BOOKING DATASET

Aim

To connect, analyze, visualize, and create interactive dashboards using **Tableau** with the **Life Expectancy Dataset** to effectively analyze life expectancy trends, country status, mortality, GDP, schooling, population, and other health-related factors across different countries and years.

Software Requirements

- Tableau Desktop / Tableau Public
- Microsoft Excel
- CSV Dataset
- Windows Operating System

Theory

Tableau is a powerful Business Intelligence (BI) and Data Visualization tool used to create interactive dashboards and reports.

The **Life Expectancy Dataset** contains information about life expectancy and various health, economic, and demographic factors, including:

- Country
- Year
- Country Status
- Life Expectancy
- Adult Mortality
- Infant Deaths
- Alcohol
- Percentage Expenditure

Tableau allows users to analyze this data and identify patterns, trends, customer behavior, and cancellation factors across different hotels and time periods.

Tableau Architecture

Life Expectancy Dataset



Data Connection



Worksheets



Visualizations



Dashboard



Story

Learning Outcomes

After completing this experiment, students will be able to:

- ✓ Connect the **Life Expectancy Dataset** to Tableau
- ✓ Create worksheets using different dimensions and measures
- ✓ Generate charts and visualizations
- ✓ Analyze life expectancy trends over time
- ✓ Compare developed and developing countries
- ✓ Apply filters and parameters
- ✓ Create calculated fields
- ✓ Analyze relationships between life expectancy and other factors
- ✓ Build interactive dashboards
- ✓ Create stories for data analysis

Sample Dataset

Life Expectancy Dataset

<i>Country</i>	<i>Year</i>	<i>Status</i>	<i>Life expectancy</i>	<i>Adult Mortality</i>	<i>GDP</i>	<i>Schooling</i>
<i>Afghanistan</i>	<i>2015</i>	<i>Developing</i>	<i>65.0</i>	<i>263</i>	<i>584.259</i>	<i>10.1</i>
<i>Afghanistan</i>	<i>2014</i>	<i>Developing</i>	<i>59.9</i>	<i>271</i>	<i>612.696</i>	<i>10.0</i>
<i>Afghanistan</i>	<i>2013</i>	<i>Developing</i>	<i>59.9</i>	<i>268</i>	<i>631.745</i>	<i>9.9</i>
<i>Afghanistan</i>	<i>2012</i>	<i>Developing</i>	<i>59.5</i>	<i>272</i>	<i>669.959</i>	<i>9.8</i>
<i>Afghanistan</i>	<i>2011</i>	<i>Developing</i>	<i>59.2</i>	<i>275</i>	<i>63.537</i>	<i>9.5</i>

The actual Life Expectancy dataset contains a much larger number of records and columns.

Algorithm 1 – Connecting Data Source

Step 1

Open **Tableau Desktop**.

Step 2

Select **Connect**.

Step 3

Choose **Microsoft Excel** or **Text File** depending on the dataset format.

Step 4

Browse and select the **Life Expectancy Dataset**.

Step 5

Open the dataset.

Step 6

Verify the imported fields such as:

- Country
- Year
- Status
- Life Expectancy
- Adult Mortality
- GDP
- Population
- Schooling

Procedure

Tableau

↓

Connect

↓

Excel / Text File

↓

Life Expectancy Dataset

↓

Open

↓

Verify Data

Output

Life Expectancy Dataset loaded into Tableau successfully.

Algorithm 2 – Creating Worksheet

Step 1

Open **Sheet 1**.

Step 2

Drag **Country** to Columns.

Step 3

Drag **Number of Records** to Rows.

Step 4

Select an appropriate visualization.

Step 5

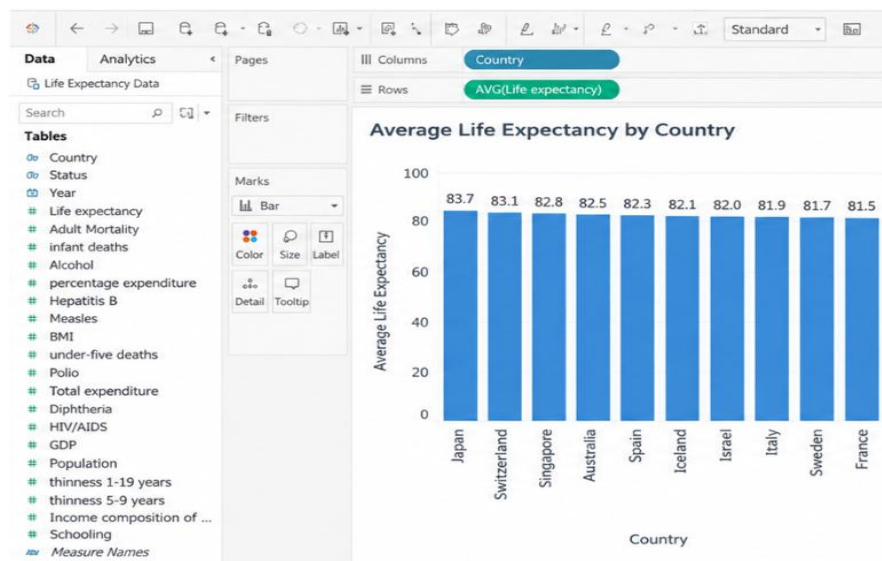
Analyze the number of life expectancy by country.

Output

Worksheet created successfully.

Interpretation

The worksheet displays the number of hotel booking recorded for different countries.



Algorithm 3 – Bar Chart Creation

Objective

To compare the **average life expectancy across different countries**.

Step 1

Drag **Country** → Columns.

Step 2

Drag **Number of Records** → Rows.

Step 3

Select **Bar Chart** from Show Me.

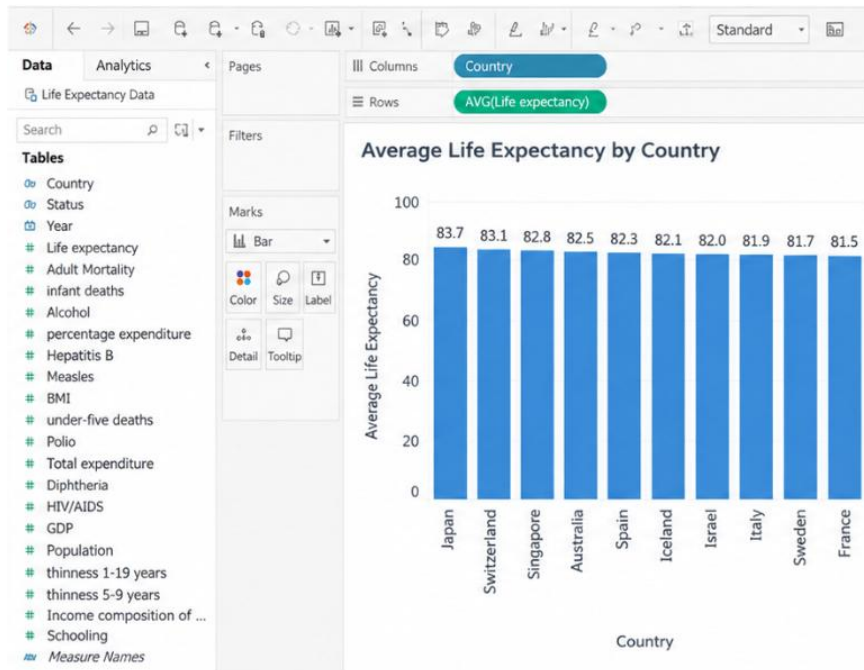
Step 4

Enable **Show Mark Labels**.

Step 5

Sort the countries in descending order.

Output Visualization



Interpretation

The bar chart helps identify life expectancy on different hotels.

Algorithm 4 – Pie Chart Creation

Objective

To analyze the distribution of hotel booking based on country name.

Step 1

Drag **Hotel Name** to the view.

Step 2

Select **Pie Chart** from Show Me.

Step 3

Drag **Number of Records** to **Angle**.

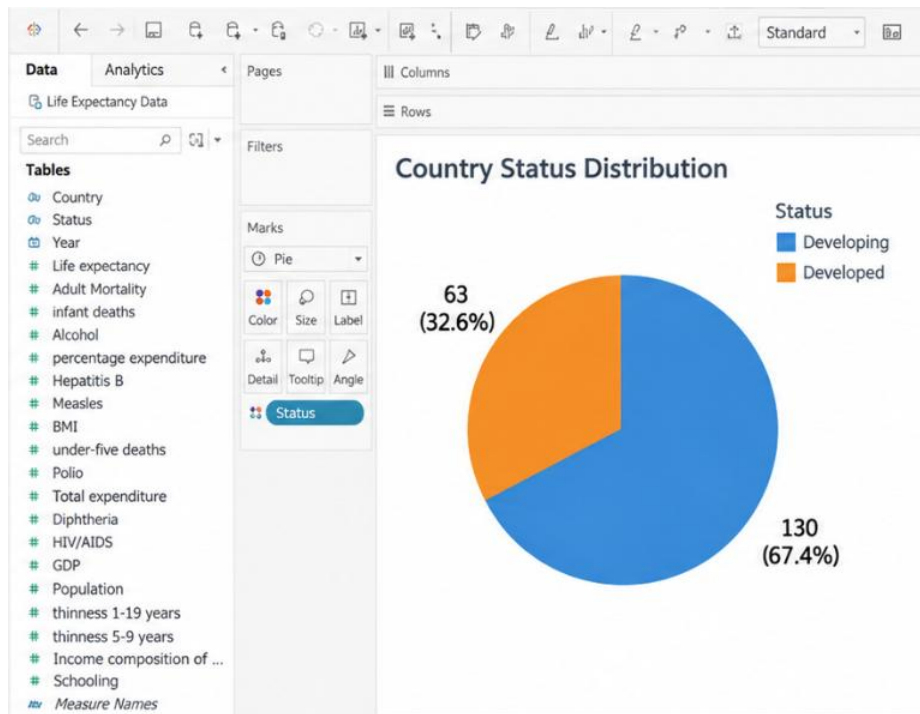
Step 4

Drag **Countries** to **Color**.

Step 5

Enable **Show Labels**.

Output Visualization



Interpretation

The pie chart shows the proportion of **Developed and Developing countries** present in the dataset.

Algorithm 5 – Line Chart Creation

Objective

To analyze the trend of **average life expectancy over the years**.

Step 1

g **Year** → Columns.

Step 2

Drag **Life expectancy** → Rows.

Step 3

Select **Line Chart**.

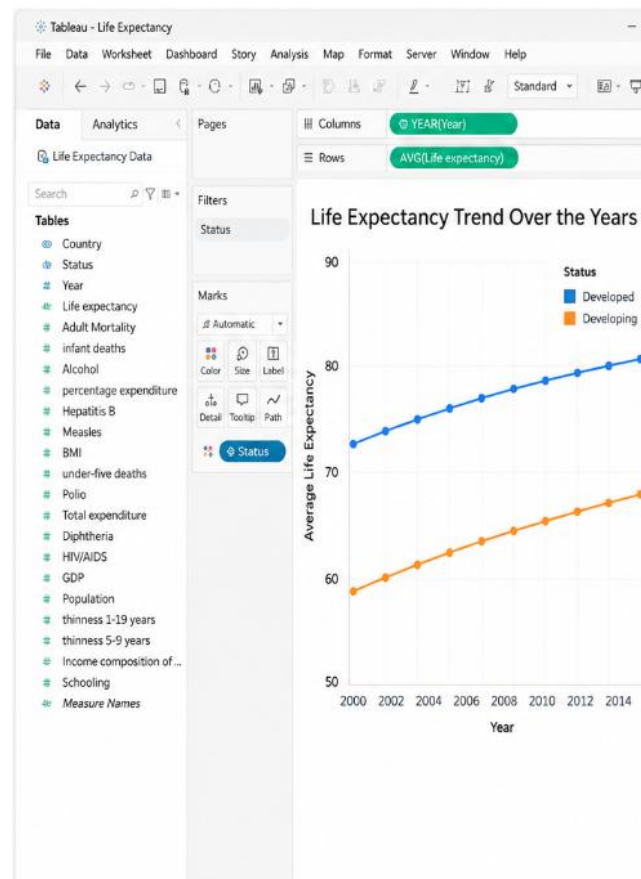
Step 4

Apply appropriate formatting.

Step 5

Analyze the trend.

Output Visualization



Interpretation

Line chart showing average life expectancy over the years. It helps identify:

- Increasing trends
- Decreasing trends
- Peak periods
- Changes in life expectancy

Algorithm 6 – Creating Calculated Field

Objective

To calculate **Life Expectancy Category** based on the life expectancy value.

Step 1

Select:

Analysis → Create Calculated Field

Step 2

Enter the field name:

Life Expectancy Category

Step 3

Enter the formula:

```
IF [Life expectancy] >= 70
```

```
THEN "High"
```

```
ELSE "Low"
```

```
END
```

Step 4

Click **OK**.

Output

Country	Life Expectancy	Life Expectancy Category
Japan	83.7	High
Afghanistan	65.0	Low
Switzerland	83.1	High
Angola	52.4	Low

Interpretation

The calculated field categorizes countries into **High** and **Low** life expectancy groups based on the selected threshold.

Algorithm 7 – Applying Filters

Objective

To analyze life expectancy for a **specific country status**.

Step 1

Drag **Status** to the **Filters** shelf.

Step 2

Select the required status.

Step 3

Apply the filter.

Step 4

Observe the updated charts.

Example

Filter: Status = Developed

Output

The visualization displays only records belonging to **Developed countries**.

Interpretation

Filters allow users to focus on a specific country status and perform detailed analysis.

Algorithm 8 – Creating Parameter

Objective

To display countries based on a user-defined **minimum life expectancy**.

Step 1

Select:

Create Parameter

Step 2

Specify the parameter name:

Minimum Life Expectancy

Step 3

Set the data type as:

Float

Step 4

Set the required minimum value.

Step 5

Create a calculated field using the parameter.

Formula

[Life expectancy] >= [Minimum Life Expectancy]

Example

Minimum Life Expectancy = 70

Output

The isualization displays countries where the **Life Expectancy is 70 years or more**.

Interpretation

Parameters allow users to dynamically control the analysis based on selected life expectancy values.

Algorithm 9 – Dashboard Creation

Objective

To create an interactive **Life Expectancy Analysis Dashboard**.

Step 1

Create multiple worksheets.

Step 2

Select **Dashboard** → **New Dashboard**.

Step 3

Drag the required worksheets into the dashboard.

Step 4

Arrange the charts.

Step 5

Add filters for:

- Year
- Country
- Status
- Life Expectancy Category

Step 6

Add appropriate titles.

Dashboard Analysis

Average Life Expectancy

Average life expectancy recorded in the dataset.

Number of Countries

Number of unique countries available in the dataset.

Average GDP

Average GDP value across the records.

Average Schooling

Average schooling value across the records.

Interpretation

The dashboard provides a consolidated view of **life expectancy, country status, economic conditions, mortality, and education**, and allows users to interactively analyze the data using filters.

Algorithm 10 – Story Creation

Objective

To present the analysis of **life expectancy, health, economic, and demographic factors** through a sequence of visual insights.

Step 1

Create the required worksheets.

Step 2

Create the dashboard.

Step 3

Select **Story** → **New Story**.

Step 4

Add the dashboard and required worksheets as story points.

Step 5

Arrange the story points in a logical sequence.

Step 6

Present the findings.



Example Story

Story Point 1

Overall Life Expectancy Analysis

Displays the overall average life expectancy, number of countries, GDP, and schooling.

Story Point 2

Year-wise Life Expectancy Trend

Shows how average life expectancy has changed over the years.

Story Point 3

Country Status Analysis

Compares life expectancy between **Developed** and **Developing** countries.

Story Point 4

Country-wise Analysis

Shows countries with higher and lower life expectancy.

Story Point 5

Health and Economic Analysis

Analyzes relationships between life expectancy and factors such as **GDP, Adult Mortality, and Schooling**.

Story Point 6

Final Conclusions

Summarizes the major patterns and observations obtained from the Life Expectancy dataset.

Applications

Healthcare Management

Analysis of life expectancy and health-related factors across countries.

Public Health Planning

Identification of countries with lower life expectancy and higher mortality.

Economic Analysis

Analysis of relationships between **GDP and life expectancy**.

Education Analysis

Understanding the relationship between **schooling and life expectancy**.

Business Intelligence

Interactive visualization and dashboard-based analysis.

Decision Making

Identification of health, education, and economic patterns for better planning.

Data Analytics

Interactive visualization and exploratory data analysis.

Result